

LILLVIKEN-ROPARUDDEN V, Sweden

WMO: 021300

Lat: 66.4860N

Lon: 16.4173E

Elev: 1578

StdP: 13.88

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-21.1	-31.1	1.0	-26.1	-25.7	1.4	-21.1	35.5	31.7	31.4	28.2	2.9	320	0.810

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	70.8	54.6	67.2	53.0	63.7	52.2	58.6	63.6	56.8	61.5	55.2	59.9	8.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.0	73.5	59.1	55.1	68.5	57.6	53.2	63.8	56.6	26.2	63.7	25.0	62.0	23.9	60.0	62.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.9	27.1	23.2	DB	-32.4	75.9	4.5	4.2	-35.6	78.9	-38.2	81.4	-40.7	83.7	-44.0	86.7	(4)
(5)			WB	-32.3	61.1	4.3	0.9	-35.3	61.8	-37.8	62.3	-40.2	62.8	-43.4	63.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.4	10.0	12.2	19.3	29.5	39.7	48.8	55.0	52.9	45.1	35.4	24.5	15.6	(6)
(7)		DBStd	18.28	14.52	15.31	11.01	7.65	5.89	6.13	4.61	4.79	5.42	6.87	10.20	15.52	(7)
(8)		HDD50	6766	1239	1059	951	616	326	97	8	24	162	452	764	1068	(8)
(9)		HDD65	11884	1704	1479	1416	1066	785	486	311	375	598	917	1214	1533	(9)
(10)		CDD50	357	0	0	0	0	5	62	163	113	14	0	0	0	(10)
(11)		CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH74	21	0	0	0	0	0	8	12	1	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	9.2	8.0	9.1	9.8	9.6	9.5	10.6	9.2	8.1	9.6	10.0	8.3	8.6	(14)
(15)	Precipitation	PrecAvg	26.3	2.3	1.8	1.7	1.2	1.5	2.5	3.6	3.0	2.6	2.2	1.9	2.2	(15)
(16)		PrecMax	34.6	3.6	4.0	2.8	2.3	2.6	4.1	5.9	6.4	5.4	3.5	3.7	4.4	(16)
(17)		PrecMin	19.1	0.7	0.4	0.6	0.4	0.4	0.5	1.1	1.2	0.7	1.0	0.6	0.9	(17)
(18)		PrecStd	4.1	0.9	0.9	0.7	0.5	0.7	1.2	1.5	1.5	1.2	0.7	0.9	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	40.1	42.5	49.2	68.7	74.0	75.5	73.5	62.0	50.5	44.7	41.5	(19)
(20)			MCWB	35.6	34.8	35.9	38.7	53.1	51.4	56.2	55.9	50.3	44.5	40.2	36.0	(20)
(21)		2%	DB	37.6	37.9	39.1	44.9	59.8	68.0	71.4	68.9	57.4	48.6	42.5	39.6	(21)
(22)			MCWB	33.1	33.6	34.5	37.6	46.5	50.5	55.4	55.5	50.4	43.2	37.9	34.7	(22)
(23)		5%	DB	34.3	35.7	36.9	42.2	54.3	63.9	68.1	64.7	54.8	46.7	39.4	37.0	(23)
(24)			MCWB	31.1	31.6	32.8	36.0	43.2	50.2	54.0	53.5	49.0	42.1	35.4	32.6	(24)
(25)		10%	DB	29.5	32.7	34.6	40.0	49.8	60.4	64.4	61.4	52.8	45.1	36.3	34.1	(25)
(26)			MCWB	27.2	29.7	30.9	34.5	41.0	49.6	53.2	53.0	47.9	41.1	33.3	30.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	36.1	36.6	40.9	54.3	57.8	60.8	60.4	55.2	46.5	42.2	37.4	(27)
(28)			MCD B	40.2	38.8	39.9	46.6	69.3	62.0	66.5	65.4	57.6	49.4	43.4	40.6	(28)
(29)		2%	WB	33.7	34.0	34.8	38.6	48.5	55.2	59.2	58.3	52.2	44.3	38.8	34.8	(29)
(30)			MCD B	37.2	37.8	38.5	43.2	56.5	60.1	65.1	62.8	54.7	47.2	41.5	38.9	(30)
(31)		5%	WB	30.8	31.8	33.0	36.9	45.0	52.7	57.5	56.4	50.7	42.9	35.6	33.0	(31)
(32)			MCD B	34.0	35.0	36.7	41.2	51.5	59.6	62.1	61.3	53.5	46.3	39.0	36.5	(32)
(33)		10%	WB	27.2	29.7	31.0	35.2	42.1	50.3	55.8	54.9	49.1	41.3	33.6	31.0	(33)
(34)			MCD B	29.5	32.7	34.2	39.6	48.0	58.6	60.4	59.1	52.2	44.7	35.7	34.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.6	16.2	16.3	14.2	12.8	13.0	13.1	12.1	9.6	7.2	8.5	12.8	(35)
(36)			MCDBR	10.6	14.8	14.0	15.3	23.4	21.0	21.1	19.1	12.9	7.9	9.1	11.3	(36)
(37)			MCWBR	8.9	12.6	10.8	9.0	11.3	8.0	7.9	8.3	7.8	5.8	6.9	8.9	(37)
(38)		5% WB	MCDBR	11.0	15.0	14.0	12.8	19.6	16.5	14.2	15.3	10.5	7.4	9.1	12.2	(38)
(39)			MCWBR	9.7	13.2	11.2	8.0	11.6	8.8	8.3	8.7	7.9	6.0	7.6	10.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.175	0.242	0.280	0.306	0.320	0.323	0.333	0.319	0.291	0.262	0.180	0.178	(40)	
(41)		taud	1.921	2.140	2.190	2.220	2.356	2.484	2.483	2.503	2.562	2.332	1.964	1.930	(41)	
(42)		Ebn at Noon	77	197	247	270	279	282	274	267	252	191	70	7	(42)	
(43)		Edh at Noon	5	16	26	33	32	29	29	25	19	13	4	1	(43)	
(44)	All-Sky Solar Radiation	RadAvg	22	169	626	1223	1565	1600	1467	1092	578	201	46	2	(44)	
(45)		RadStd	1	28	47	72	129	115	173	111	64	30	5	0	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page